

EXPLORE · INNOVATE · TRANSFORM



Smart & Accessible Navigation for KSR Bengaluru Railway Station

Navigate. Discover. Access.

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Idea Title

The Problem → Our Solution

RailMarga— Smart & Accessible KSR Navigation

PROBLEM

Finding your way inside KSR is harder than it looks.



First-time passengers struggle to locate platforms and facilities



Important locations such as lifts, toilets, exits and ticket areas are difficult to discover



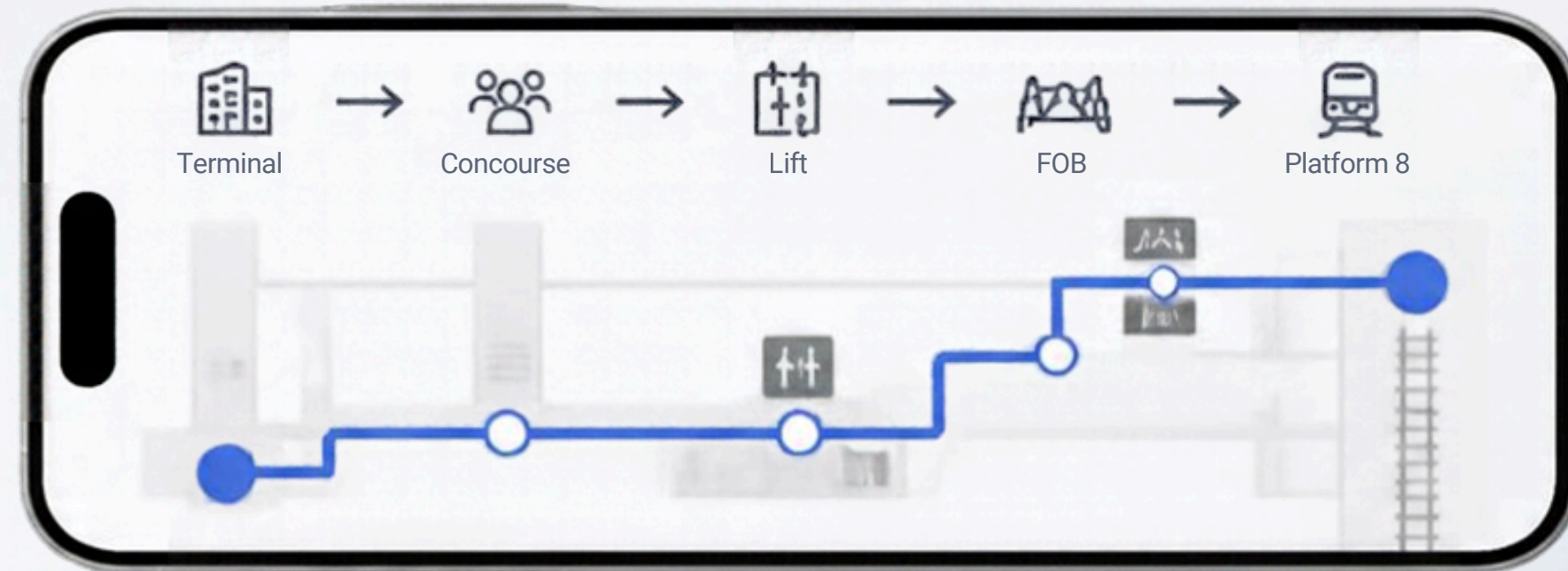
Wheelchair and elderly passengers may need different routes



Blocked or unavailable paths can make a planned route unusable

SOLUTION

RailMarga turns the station map into personalized navigation.



Personalized Navigation

Routes based on passenger needs



Accessibility-Aware Routing

Prefer lifts, ramps and accessible paths



Dynamic Rerouting

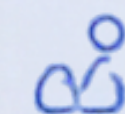
Automatically avoid blocked paths

Why is it different?



Static station map

shows WHERE things are



RailMarga

tells passengers HOW to reach them

Technical Approach

1 TECHNICAL APPROACH

1 KSR Station Map



SVG-based
station layout

2 Map Important Locations



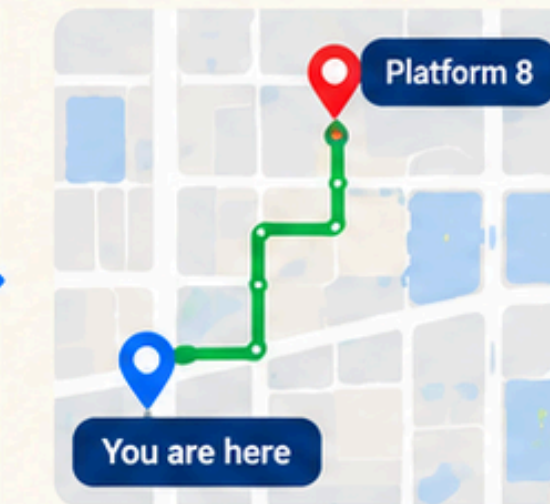
Platforms • lifts • stairs
• ramps • facilities

3 Build Navigation Graph



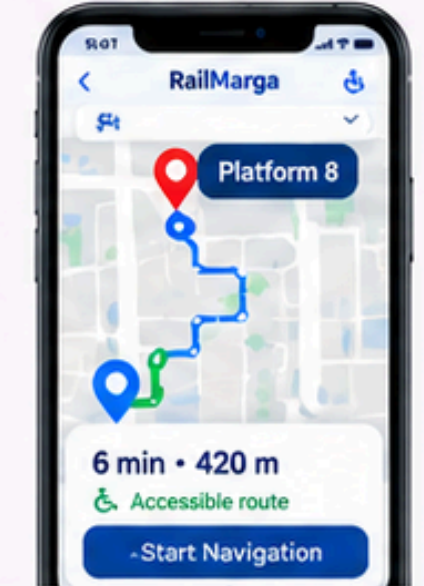
Nodes + walkable paths

4 Calculate Best Route



A* / Dijkstra

5 Mobile Navigation



Route + visual +
voice guidance

2 TECHNOLOGY STACK



React Native
Mobile App



TypeScript + Expo
Development



SVG +
react-native-svg
Interactive Map



Node.js + Express
Backend API



MongoDB
Station Data



Text-to-Speech
Voice Guidance



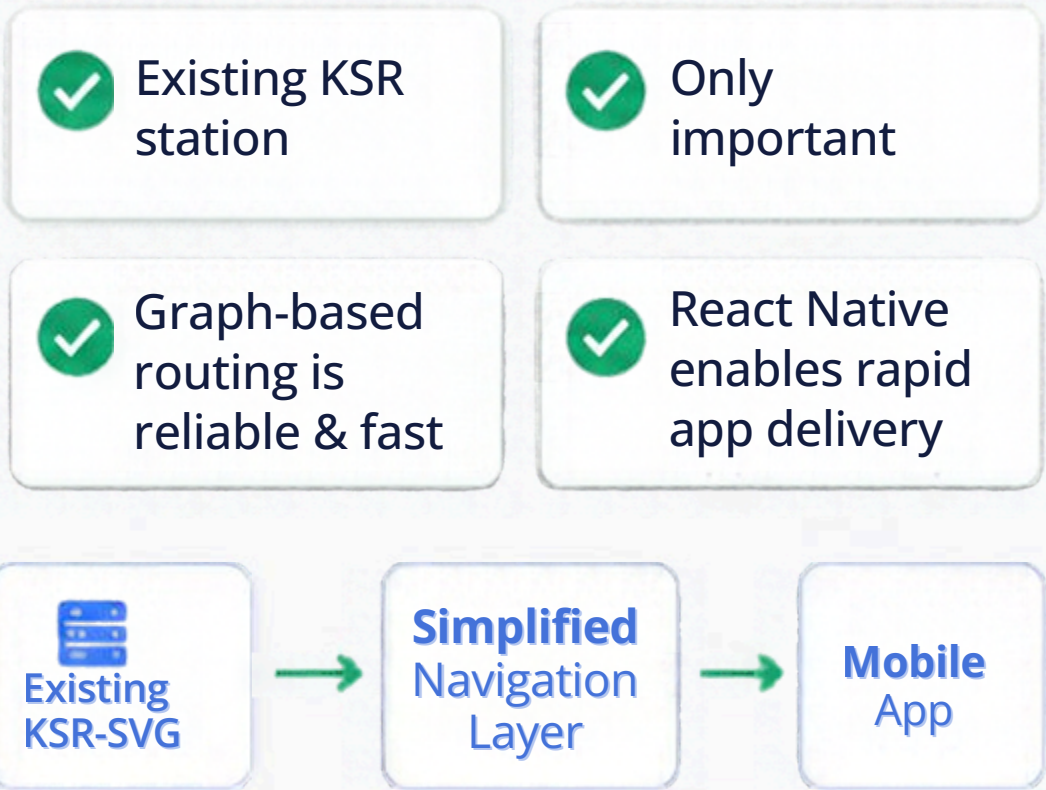
QR Checkpoints
Indoor Positioning



Feasibility & Viability



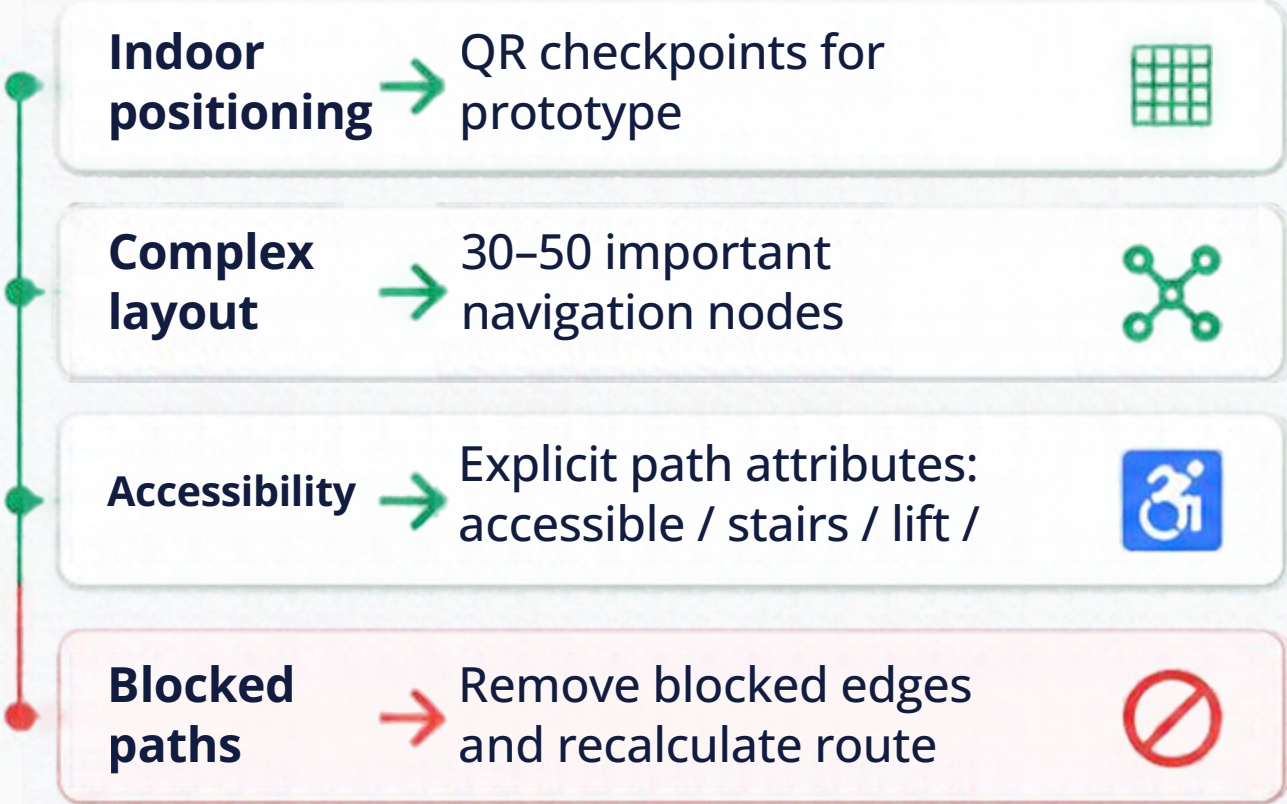
WHY IT IS FEASIBLE



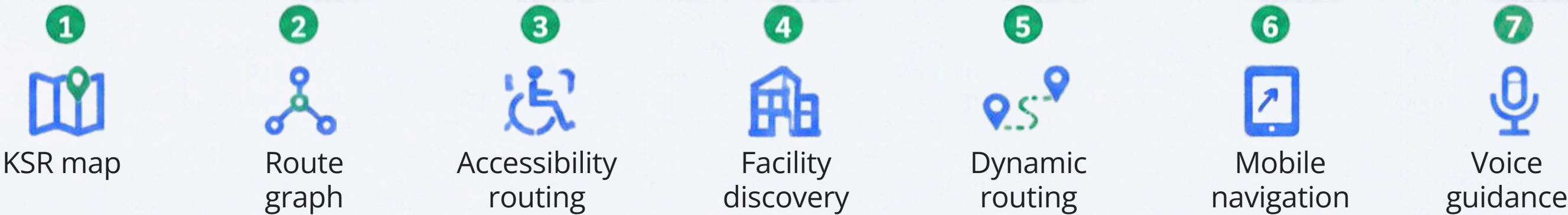
CHALLENGES



OUR STRATEGY



MVP



Impact & Benefits

Making KSR easier to navigate for everyone



FIRST-TIME PASSENGERS

Clear step-by-step directions reduce confusion inside an unfamiliar station.



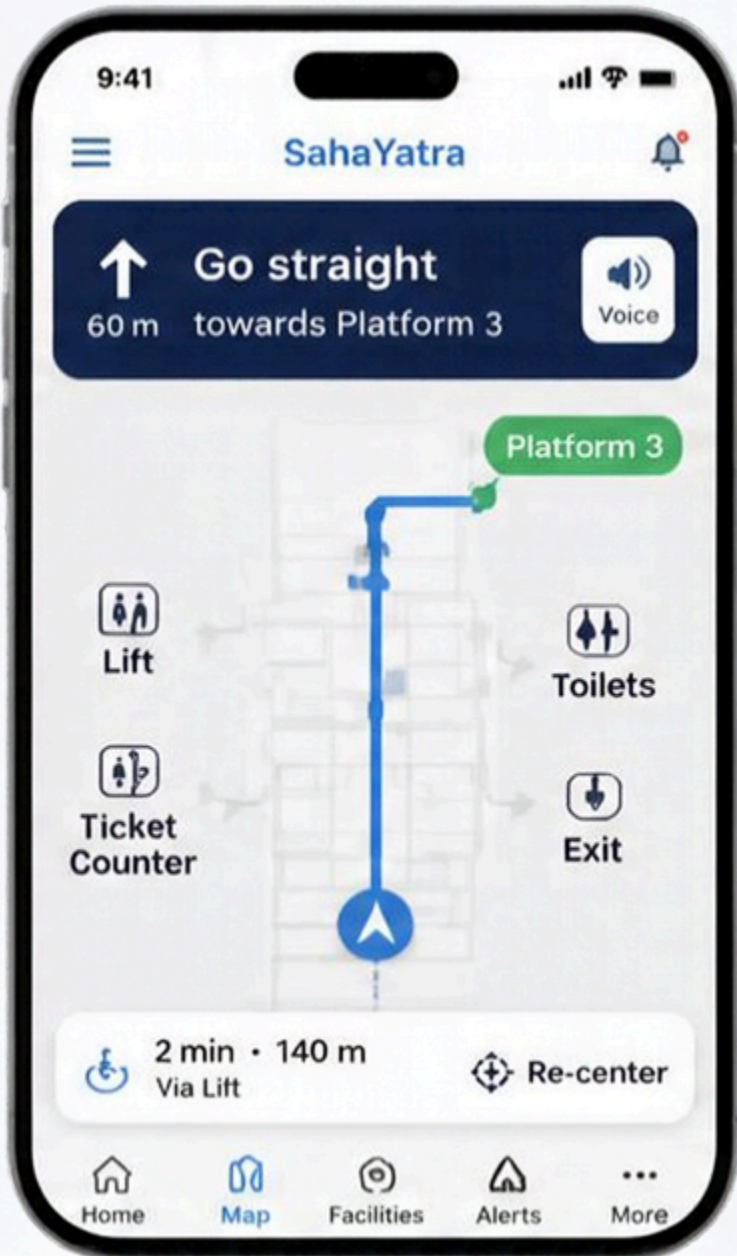
ELDERLY PASSENGERS

Routes can prioritize lifts and reduce difficult walking or stair sections.



WHEELCHAIR USERS

Accessibility-aware routes prioritize usable paths, lifts and ramps.



ALL PASSENGERS

Quick discovery of platforms, toilets, exits, ticket counters and other facilities.



STATION OPERATIONS

Blocked paths can trigger alternative routes instead of leaving passengers stranded.



CONFUSION



PERSONALIZED GUIDANCE



ACCESSIBLE ROUTE



BETTER PASSENGER EXPERIENCE



Future Potential

Extend the same navigation framework to other railway stations.

